

Pest Patrol Drone Design Brief

NAME _____

DATE _____

PERIOD _____

Part 1: Plan from evidence

Design a drone concept that responds to the three field reports in *Find Your Future*. You are designing the **idea and evidence chain**, not a working aircraft.

REQUIRED DESIGN EVIDENCE

- ☐ At least six labeled features
- ☐ A sensor or camera and what it detects
- ☐ A movement or field-coverage plan
- ☐ A way to report information to the farmer
- ☐ A safety or crop-protection feature
- ☐ Three labels that point back to field-report evidence
- ☐ One tradeoff the design cannot completely avoid

COPY YOUR THREE CONSTRAINTS

Constraint 1:

Constraint 2:

Constraint 3:

FEATURE PLAN

Feature I will label first: _____

It responds to this evidence:

Build route: ☐ Paper and pencil ☐ Canva ☐ Adobe Express ☐ Other approved route

Part 2: Draw and label the drone

Use the full page. Labels may sit around the drawing with arrows pointing to each feature. Rough drawings are acceptable when the engineering idea is clear.



Part 3: Explain one design decision and one tradeoff

EVIDENCE CHAIN

The field report says:

So my drone includes:

This feature would help because:

TRADEOFF

One benefit of my design:

One cost, risk, or limit:

My team should keep/change this feature because:

SUBMISSION CHECK

- ☐ Six or more features are labeled, and I submitted the draft in Canvas or on paper.
- ☐ Three labels connect to field-report evidence.
- ☐ My explanation follows evidence -> feature -> benefit.
- ☐ I named a real tradeoff instead of claiming the design is perfect.